

Unsupervised learning

Unsupervised learning

- ▶ We only observe $\mathbf{x}$, not y
- ▶ Exploratory / descriptive data analysis
- ▶ Detect what is unusual/remarkable with some observations
- ▶

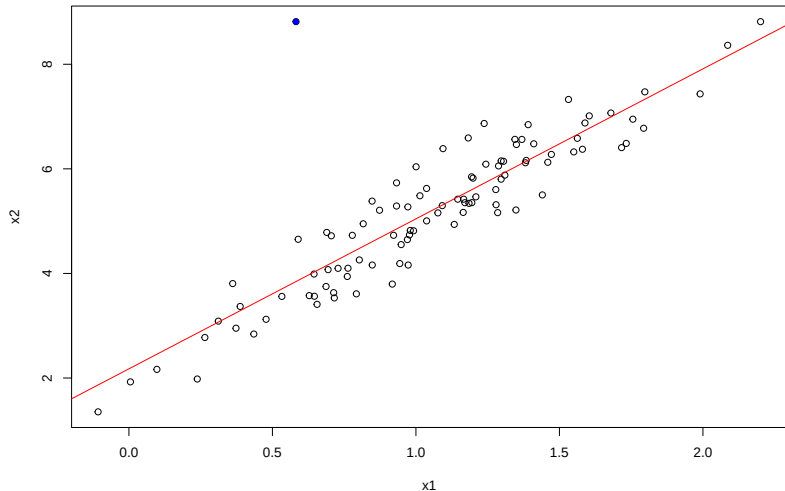


Speculate / hypothesise on why they are unusual/remarkable

- ▶ Hypotheses generating methods
- ▶ “Advanced descriptive analysis”
- ▶ Principal component analysis (PCA) and cluster analysis

Principal component analysis

Visualise two x -variables



What if $p = 10$

- ▶ $\binom{10}{2} = 45$ such plots?
- ▶ How can we summarise the 10 variables so that we get two variables carrying as much of the variables $x_1, x_2, \dots, x_{10}$ as possible?

The first principal component

► Let $Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p$

►

$$\text{Var}(Z_1) = \sum_{i=1}^p \phi_{i1}^2 \text{Var}(X_i) + 2 \sum_{i \neq j} \phi_{i1} \phi_{j1} \text{Cov}(X_i, X_j)$$

► Find the *loadings* $\phi_{11}, \phi_{21}, \dots, \phi_{p1}$ that solves

$$\begin{cases} \text{maximize } \text{Var}(Z_1) \\ \phi_{11}, \phi_{21}, \dots, \phi_{p1} \\ \text{subject to} \\ \sum_{j=1}^p \phi_{j1}^2 = 1 \end{cases}$$

► This is the *first principal component*

The second principal component

- ▶ Let $Z_2 = \phi_{12}X_1 + \phi_{22}X_2 + \dots + \phi_{p2}X_p$



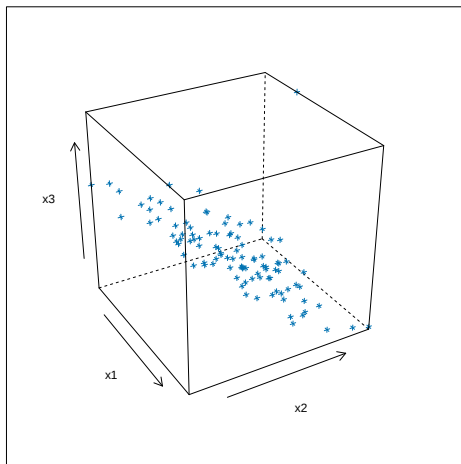
$$\text{Var}(Z_2) = \sum_{i=1}^p \phi_{i2}^2 \text{Var}(X_i) + 2 \sum_{i \neq j} \phi_{i2} \phi_{j2} \text{Cov}(X_i, X_j)$$

- ▶ Find the *loadings* $\phi_{12}, \phi_{22}, \dots, \phi_{p2}$ that solves

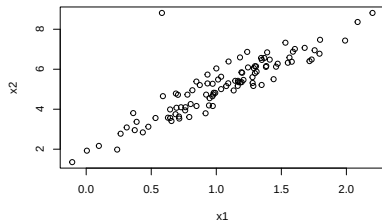
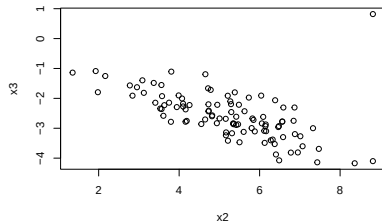
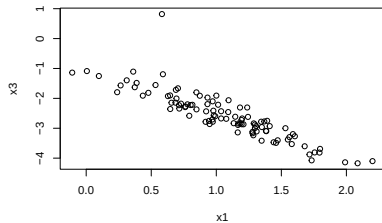
$$\left\{ \begin{array}{l} \text{maximize } \text{Var}(Z_2) \\ \phi_{12}, \phi_{22}, \dots, \phi_{p2} \\ \text{subject to} \\ \sum_{j=1}^p \phi_{j2}^2 = 1 \\ \text{Cov}(Z_1, Z_2) = 0 \end{array} \right.$$

- ▶ This is the *second principal component*. Among the linear combination of the X 's, uncorrelated with Z_1 (the first principal component), Z_2 has largest variance.

$p=3$, reduce to two dimensions



$p=3$, reduce to two dimensions



p=3, reduce to two dimensions

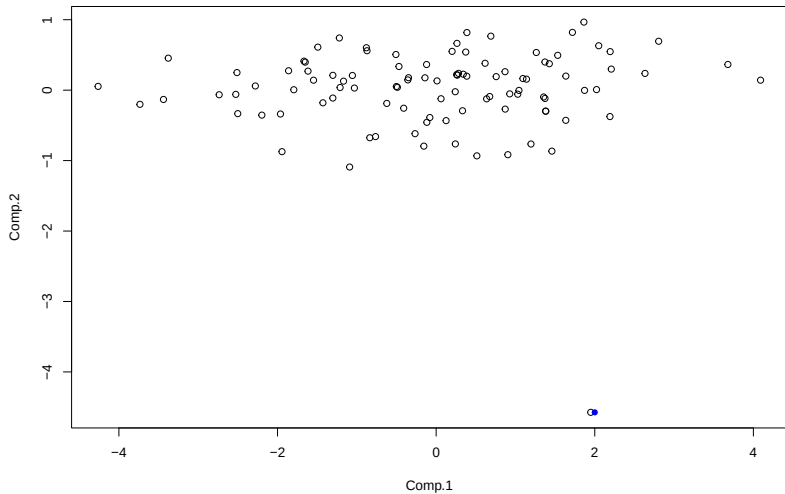
```
pca2=princomp(x)  
summary(pca2)
```

Importance of components:

	Comp.1	Comp.2	Comp.3
Standard deviation	1.5940343	0.6333490	0.031345120
Proportion of Variance	0.8633688	0.1362973	0.000333842
Cumulative Proportion	0.8633688	0.9996662	1.000000000

Almost all variation in the three variables can be summarised in two linear combinations.

$p=3$, reduce to two dimensions



USArrest data

```
require(ISLR)
nrow(USArrests)
```

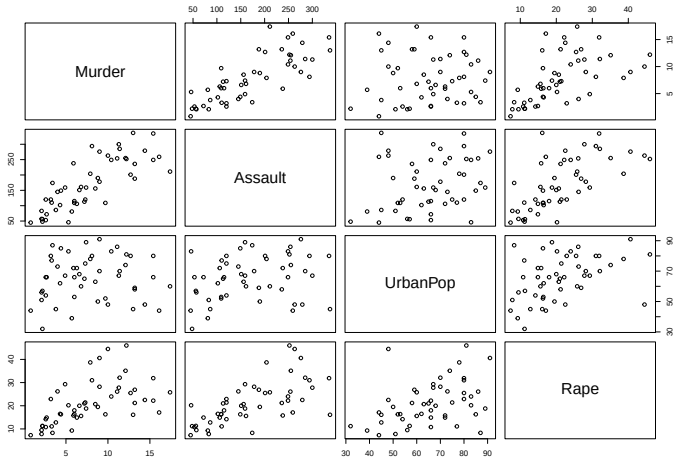
```
[1] 50
```

```
round(cor(USArrests),3)
```

	Murder	Assault	UrbanPop	Rape
Murder	1.000	0.802	0.070	0.564
Assault	0.802	1.000	0.259	0.665
UrbanPop	0.070	0.259	1.000	0.411
Rape	0.564	0.665	0.411	1.000

USArrest data

```
pairs(USArrests)
```



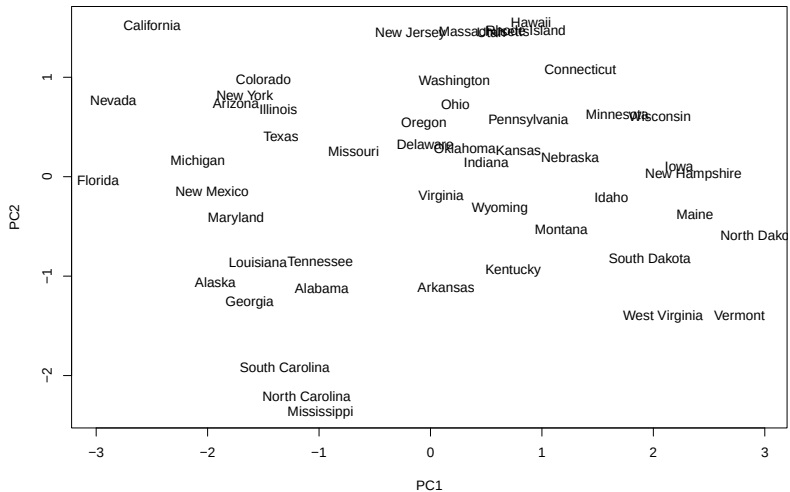
USArrest data

```
pca3=prcomp(USArrests,scale=TRUE)
summary(pca3)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	1.5749	0.9949	0.59713	0.41645
Proportion of Variance	0.6201	0.2474	0.08914	0.04336
Cumulative Proportion	0.6201	0.8675	0.95664	1.00000

USArrest data



USArrest data - loadings

```
pca3$rotation
```

	PC1	PC2	PC3	PC4
Murder	-0.5358995	-0.4181809	0.3412327	0.64922780
Assault	-0.5831836	-0.1879856	0.2681484	-0.74340748
UrbanPop	-0.2781909	0.8728062	0.3780158	0.13387773
Rape	-0.5434321	0.1673186	-0.8177779	0.08902432

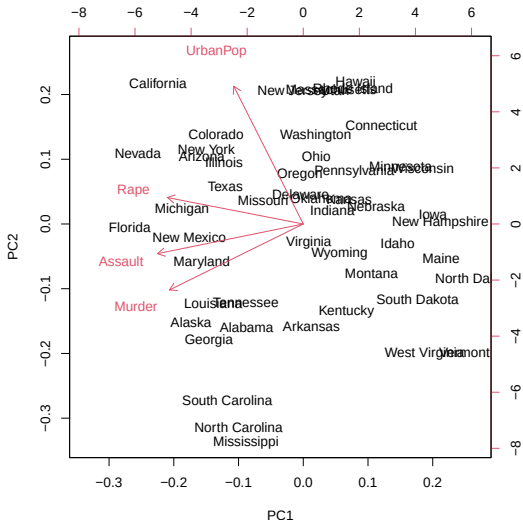
```
sum(pca3$rotation[,1]^2)
```

```
[1] 1
```

- ▶ A state with a small number on PC1 has a high rate of the crime variables.
- ▶ The crime variables have been summarized in PC1
- ▶ A state with a small number on PC2 has a high level of urbanisation.

USArrest data

```
biplot(pca3)
```



Clustering methods

- ▶ Find homogeneous subgroups among observations; find observations that are similar.
- ▶ We need to define what we mean by “similar”. This is context dependent.
- ▶ Example 1: Letting a number of features of plants determine how to classify them.
- ▶ Example 2: Using previous patterns of shopping to identify subgroups of consumers that might be more inclined to buy another product.

K-means clustering

- ▶ Predetermined number of clusters: K
- ▶ $C_1, C_2, \dots, C_K$ are sets containing the indices from the different observations $\{1, \dots, n\}$
- ▶ The union of the K subset represents all observations

$$C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$$

- ▶ The intersection of two clusters is the empty set

$$C_k \cap C_{k'} = \emptyset$$

- ▶ Thus, all observations belong to exactly one cluster.
- ▶ If observation nr i belong to the k 'th cluster, then $i \in C_k$.

K-means clustering

- ▶ Find cluster such that the within-cluster variation is small.
- ▶ Let $W(C_k)$ be a measure of within-cluster variation



$$\text{minimise}_{C_1, C_2, \dots, C_K} \left\{ \sum_{k=1}^K W(C_k) \right\}$$

- ▶ The squared Euclidian distance is a common choice of measuring within-cluster variation.

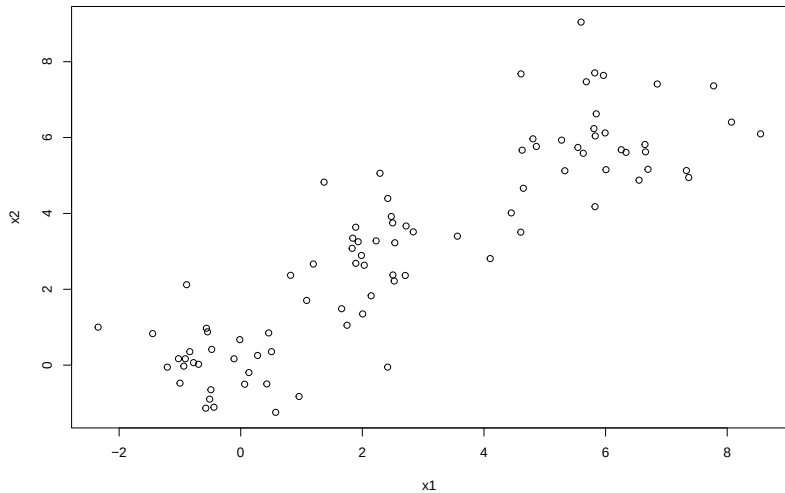
$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2$$

- ▶ A computationally demanding optimisation problem.
- ▶ We have a good, if not completely optimal, solution to this.

K-means clustering algorithm

1. Allocate the observations randomly to a cluster
2. Iterate the following until convergence
 - a. For each cluster, compute the cluster *centroid*, i.e. the mean of the x-variables in that cluster.
 - b. Assign each observation to the cluster with the closest centroid.

An example with two variables and 3 clusters



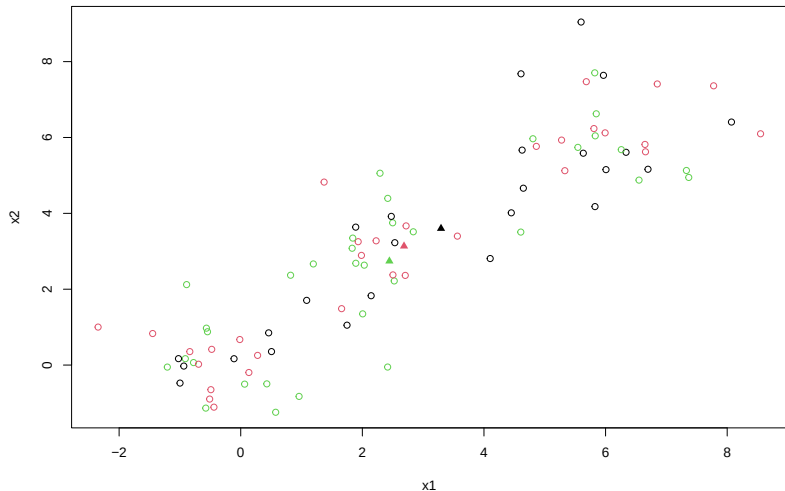
An example with two variables and 3 clusters

First step, allocate randomly to three clusters

```
K=3
farg=c("black","red","blue")
ind=sample(1:K,size=nobs,replace=TRUE)
plot(x,col=farg[ind])
m=matrix(0,K,ncol(x))
for(k in 1:K)
{
  m[k,]=sapply(x[ind==k,],mean)
  points(m[k,1],m[k,2],pch=17,col=farg[k])
}
```

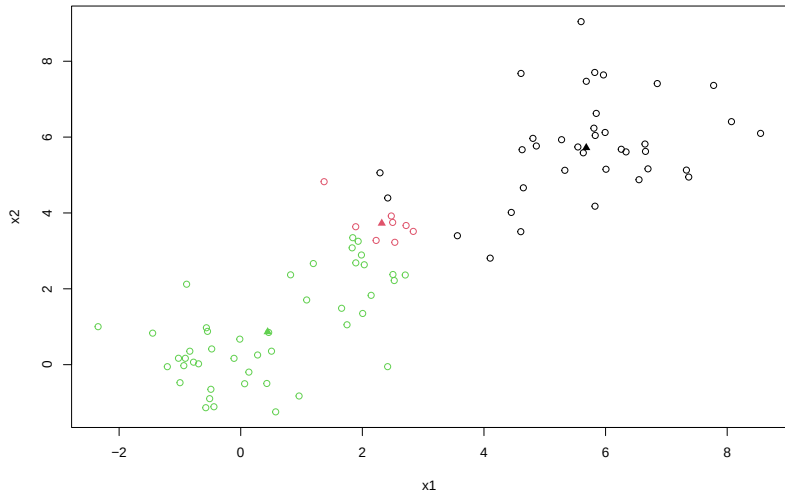
An example with two variables and 3 clusters

First step, allocate randomly to three clusters



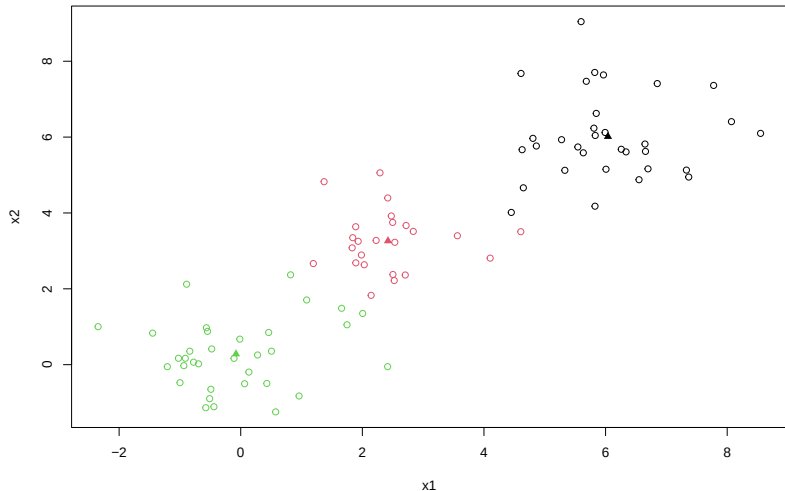
An example with two variables and 3 clusters

Second iteration



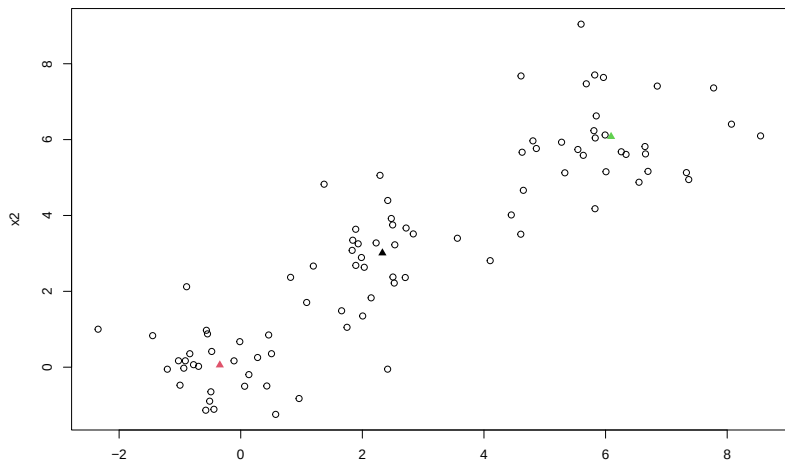
An example with two variables and 3 clusters

Third iteration



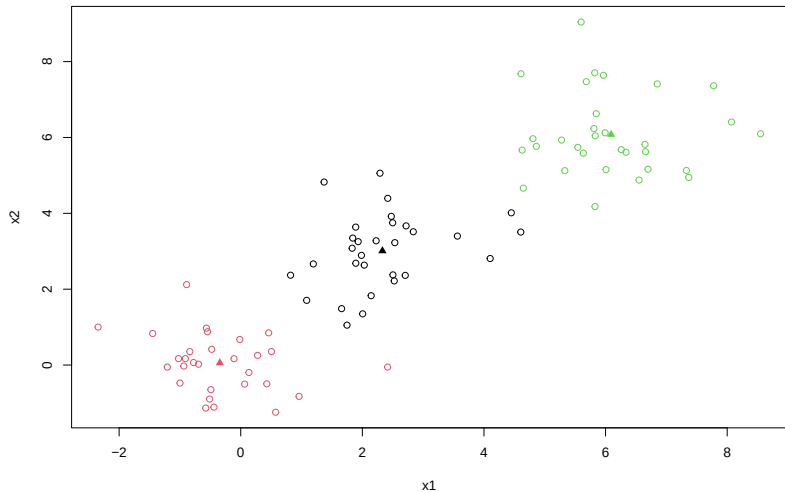
Built-in function

```
plot(x)  
km3=kmeans(x,3)  
points(km3$centers,pch=17,col=farg)
```



Built-in function

```
plot(x,col=km3$cluster)  
points(km3$centers,pch=17,col=farg)
```



Hierarchical clustering

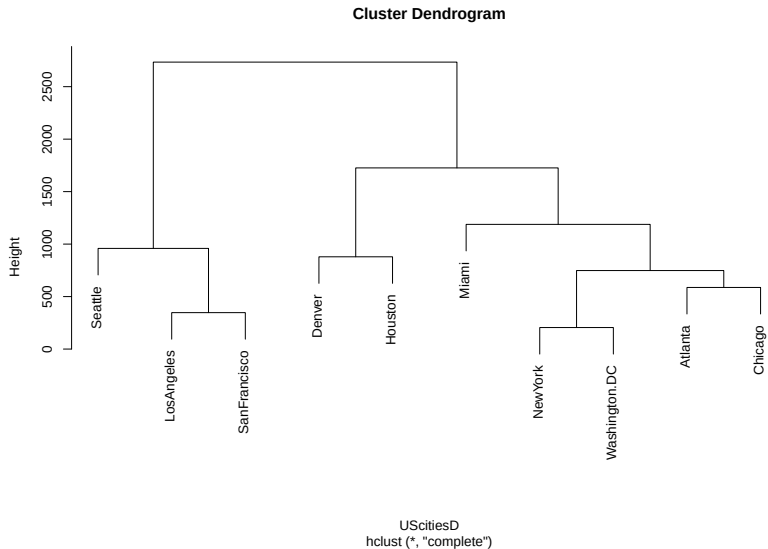
- ▶ In contrast to K-means clustering, we do not have to pre-specify the number of clusters.
- ▶ Furthermore, we obtain a tree-based representation, a *dendrogram*, of the result
- ▶ Bottom-up (agglomerate) approach

Hierarchical clustering

Start with a distance matrix (see the example section of the help to the `hclust`-function in R)

	Atlanta	Chicago	Denver	Houston	LosAngeles
Atlanta	0	587	1212	701	1936
Chicago	587	0	920	940	1745
Denver	1212	920	0	879	831
Houston	701	940	879	0	1374
LosAngeles	1936	1745	831	1374	0

Hierarchical clustering



Hierarchical clustering

- ▶ If you would like to divide the US into small regions, you “cut the tree” close to the leaves (cities). Example: Los Angeles and San Francisco (Los Francisco) can be merged into one region while Seattle remains a region on its own.
- ▶ If you would like to create large regions cut the tree further up. Example: Seattle is merged with Los Francisco, Denver with Houston (Denston) and the rest to one region (Therest).

The algorithm (ISLR, page 395)

1. We have n observations and have measured $n(n - 1)/2$ distances/dissimilarities between. To start with, each observation is its own cluster.
 2. For $i = n, n - 1, \dots, 2$:
 - a) Check through all pairwise distances between clusters and determine which pair have the smallest distance and fuse these two clusters. This distance will be the height of the dendrogram where the fusion is placed.
 - b) Compute the new pairwise inter-cluster distances among the $i - 1$ remaining pairs.
- There are a few options, so called *linkages*, on how to compute the pairwise inter-cluster distances.

Linkages

- ▶ Complete: We compute all possible distances between observations in the two clusters and choose the maximum.
- ▶ Single: As in complete but choose the minimum distance.
- ▶ Average: As in complete but compute the average of all possible distances.
- ▶ Centroid: Compute the centroid for each cluster (compare with K-means clustering) and compute the distance between those.